Does AI Widen Employment Gaps?

Tracking Early-Career Employment by Occupational Exposure in Norway

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August 2026, *Preliminary*

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Motivation

AI and the future of work: in the headlines

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Recent literature: AI exposure and employment

Country	Paper	Headline (youngest workers, top quintile)
United States	Brynjolfsson, Chandar & Chen (2025)	Q5 ages 22–25 decline $\sim$ 15 log pts
Sweden	Lodefalk et al. (2026)	Q5 ages 22–25 decline $\sim$ 5.5 %; monotonic gradient
United Kingdom	Teeselink (2025)	-0.3% per SD LLM (large language model) exposure; concentrated in junior, high-wage
39 countries	Teeselink (2026)	Hiring declines; sharper under strict employment protection legislation
Denmark	Humlum & Vestergaard (2026)	Null: early-career decline not driven by firm AI adoption; rules out effects $>$ 2 %
Finland	Kauhanen & Rouvinen (2026)	Null; trends reflect demographics, not AI

Contributions

- A public monthly dashboard, **kiindeksen.no**: employment in the universe of Norwegian private-sector payroll jobs, tracked by AI exposure with roughly a three-month data lag.
- Test whether the "canary in the coal mine" pattern (Brynjolfsson et al.) generalizes: selected high-exposure occupations vs. the **full exposure distribution**.
- Study the recent impact of **Agentic AI** ("*autonomous LLM-based systems that plan, use tools, and execute multi-step research tasks*", Korinek 2025).

Data

Data: A-ordningen

- Norway's mandatory employer-reporting system; universe of formal employment.
- Coverage: $\sim$ 3.1 million employment records per month, January 2021 through April 2026.
- Outcome: **paid payroll employment**, active relationships with positive cash earnings.
- Variables: 4-digit STYRK-08 occupation, age (monthly precision), sector, cash earnings (*kontantlønn*), position percentage, new-job flag.
- Decade age groups: 21–30 (early career), 31–40, 41–50, 51–60 (senior).
- **Private sector only**: the same quintile contains different work in the public sector.
- Seasonally adjusted (fixed calendar-month factors, 2021–2024); indexed to October 2022 $= 1$, the month before ChatGPT's public release.

Exposure measures and quintiles

- Main measure: Eloundou et al. (2024) GPT-4 exposure score β; maps 397 of 407 STYRK-08 codes (97.5 % of employment).
- Secondary: Handa et al. (2025) revealed Claude usage, split into **automation** and **augmentation**; maps 352 codes (86.5 %).
- Rank the mapped codes by exposure; assign each 4-digit code to one of five quintiles on the *occupation* distribution.
- Each code counts once, regardless of its employment share. Assignment is fixed throughout the period.

Context: Norway ranks 3rd worldwide in population-level AI use (Microsoft 2026); enterprise adoption grew from 11 % of firms in 2021 to 30 % in 2025 (SSB).

Focus on exposure measure of Eloundou et al.

- **Commonly used, standard measure.** Published in *Science* (2024), used by Brynjolfsson, Chandar & Chen (2025), later by Teeselink (2025), Humlum & Vestergaard (2026), Kauhanen & Rouvinen (2026), Lodefalk et al. (2026).
- **Best validation against observed AI use.** In Lindenlaub, Oh, Rodríguez & Veldkamp (2026, Fig. 1), Eloundou et al. has $R^2 = 0.30$ against worker-reported adoption, ahead of Handa (0.26), Felten (0.26), and Webb (0.02).

Show Lindenlaub et al. Fig. 1

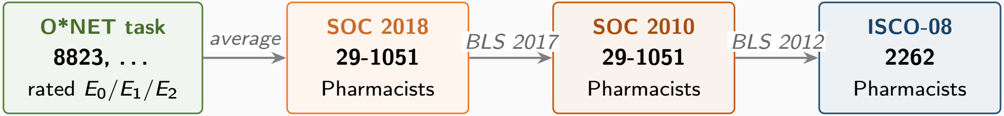
Mapping from US to Norwegian occupation codes

- Eloundou et al.'s exposure measure is built on US occupation codes (SOC).
- Norwegian register data use STYRK-08.
- STYRK-08 is based on ISCO-08; overlapping 4-digit unit groups are code-compatible after a small set of documented Norwegian-title checks.
- Bridge: SOC $\rightarrow$ ISCO-08 via US Bureau of Labor Statistics (BLS), then match overlapping STYRK-08 codes with manual SOC-source corrections for 2267 and 2269.

What Eloundou et al. measure

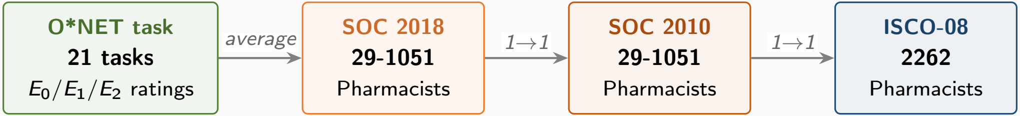
- Each O*NET task rated by humans and GPT-4 on three categories:
  - E_0 no exposure
  - E_1 LLM alone reduces task time by $\geq 50\%$
  - E_2 LLM with complementary software reduces task time by $\geq 50\%$
- Task-level score: $\beta_{\text{task}} = 0$ if E_0, 1 if E_1, 0.5 if E_2
- Occupation score: β_{occ} = mean of β_{task} with O*NET Core tasks weighted 2, Supplemental tasks 1
- We use the published occupation-level file (`occ_level.csv`); it differs slightly from unweighted task means where an occupation has Supplemental tasks

The code chain



Overlapping 4-digit unit groups in STYRK-08 are code-compatible with ISCO-08 (SSB Notater 17/2011). STYRK 2262 is Farmasøyter.

Worked example: Pharmacists (one-to-one mapping)



- Pharmacists carry the same 6-digit SOC in both vintages: 29-1051
- BLS crosswalk sends 29-1051 to a single ISCO-08 code (2262)
- Norwegian STYRK 2262 is Farmasøyter, no Norway-specific re-coding

Task-level scoring for Pharmacists (SOC 29-1051, all 21 tasks)

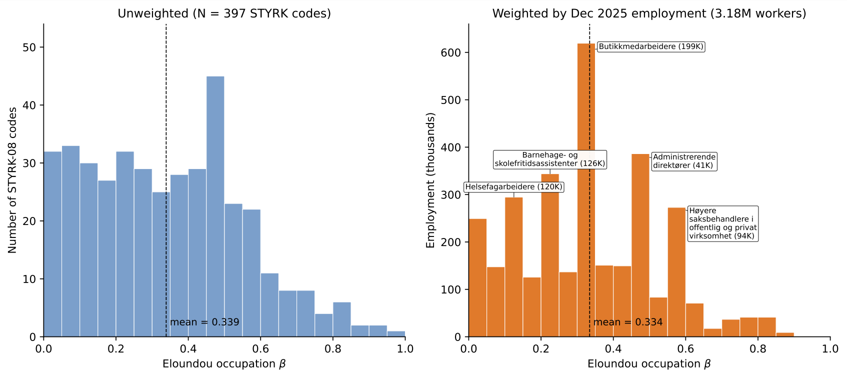
O*NET task	E	β
Review prescriptions for accuracy	E_2	0.5
Assess identity, strength, purity of meds	E_0	0.0
Provide info on drug interactions	E_2	0.5
Analyze prescribing trends	E_2	0.5
Maintain pharmacy files and patient profiles	E_2	0.5
Collaborate with other health professionals	E_2	0.5
Plan procedures for mixing and labeling	E_0	0.0
Order and purchase pharmaceutical supplies	E_2	0.5
Compound and dispense medications	E_0	0.0
Contact insurance companies on billing	E_2	0.5
Advise customers on medication brands	E_2	0.5
Teach pharmacy students serving as interns	E_1	1.0
Provide services for diabetes, asthma, etc.	E_2	0.5

O*NET task	E	β
Refer patients to other professionals	E_2	0.5
Work in hospitals/clinics or for HMOs	E_2	0.5
Update or troubleshoot pharmacy databases	E_1	1.0
Manage pharmacy operations, hire, supervise	E_2	0.5
Prepare sterile solutions	E_0	0.0
Offer health promotion / prevention activities	E_0	0.0
Assay radiopharmaceuticals	E_0	0.0
Publish educational information	E_1	1.0

Counts: 6 E_0 3 E_1 12 E_2

Occupation β (mean of 21): 0.4286 (Q4)

Distribution of Eloundou et al. β across Norwegian occupations



Left: one observation per STYRK-08 code (397 codes). Right: weighted by December 2025 employment (3.19M workers); five tallest spikes labelled. Means nearly identical; weighted mass concentrates in a handful of large mid-skill occupations.

Five largest occupations per exposure quintile (Apr 2026)

Occupation	Score	Emp.
Q1 (least exposed)		
Cleaners (offices, hotels, . . .)	0.03	40k
Building electricians	0.13	26k
Food machine operators	0.09	23k
Motor vehicle mechanics	0.09	17k
Plumbers and pipe fitters	0.06	17k
Q2		
Carpenters and joiners	0.14	35k
Child care workers	0.25	32k
Waiters	0.16	20k
Machinery mechanics and repairers	0.13	16k
Cooks	0.19	16k
Q3		
Shop sales assistants	0.34	115k
Stock clerks	0.33	29k
Heavy truck and lorry drivers	0.31	21k
Science technicians (n.e.c.)	0.36	20k
Car, taxi and van drivers	0.28	14k

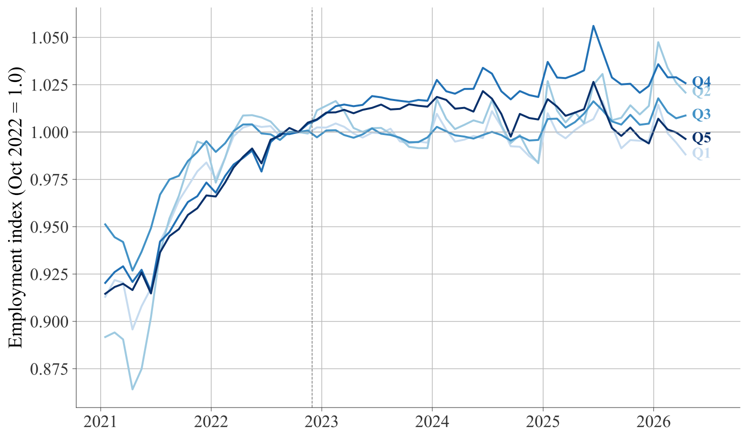
Occupation	Score	Emp.
Q4		
Retail/wholesale trade managers	0.48	30k
Managing directors / CEOs	0.49	28k
Civil engineers	0.48	19k
Sales and marketing managers	0.48	13k
Construction managers	0.44	12k
Q5 (most exposed)		
Commercial sales representatives	0.57	37k
General office clerks	0.60	33k
Systems analysts	0.63	22k
Software developers (n.e.c.)	0.77	18k
Accounting associate profs.	0.80	15k

Employment in thousands. Private sector, ages 21–60, ranked within quintile by April 2026 headcount.

More worked examples

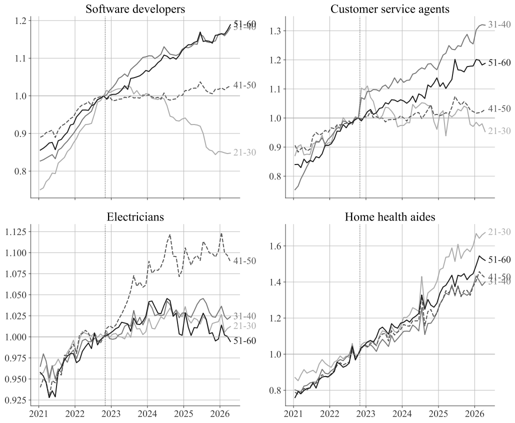
Employment gaps by AI exposure

All ages pooled: the most exposed do not pull away



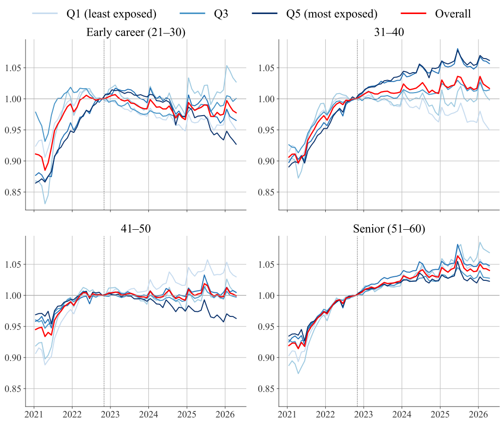
Ages 21–60, private sector, seasonally adjusted, indexed to October 2022 = 1. Q5 (most exposed) tracks Q1 (least exposed) throughout. The paper analog of the kiindeksen.no headline figure.

Case studies: the canary pattern is real



Software developers aged 21–30 decline sharply after ChatGPT ($\sim -18\,\%$ per capita); customer service agents show the same shape, milder. Electricians and home health aides, the low-exposure benchmarks, do not.

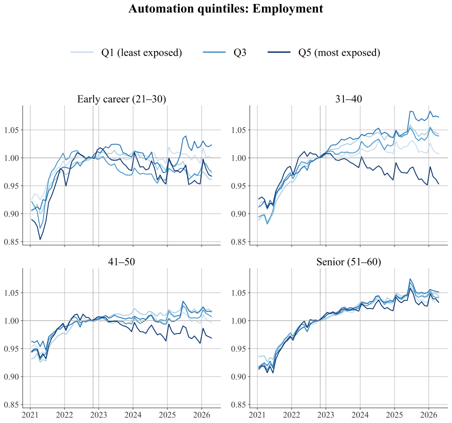
Private sector: employment by exposure quintile and age



Seasonally adjusted. No clear pattern of Q5 losing ground among the youngest: at 21–30, Q5 drifts a few points below Q1 from mid-2025, but the gap is smaller than pre-period swings.

Public sector
Cash earnings

Automation vs. augmentation (Handa et al.)



Quintiles ranked by **automation** share of Claude usage. Among the two youngest groups, the most-exposed quintile declines relative to the least exposed, more visibly from late 2025. The augmentation ranking shows no such gradient.

Augmentation

Regression analysis

Empirical specification

- Cells: (occupation j $\times$ age group a $\times$ month t), private sector, 2021m1–2026m4
- One Poisson PPML regression per decade age group:

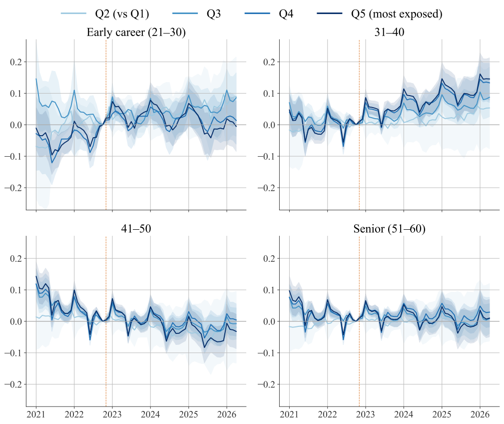
$$\log E[\text{count}_{j,a,t}] = \alpha_j + \beta_t + \gamma_{q(j),\,k(t)}$$

- α_j: occupation FE, absorbs permanent occupation size
- β_t: month FE, absorbs the aggregate age-cohort time path
- $\gamma_{q,k}$: log-point change for quintile q at event time k, relative to **Q1** (least exposed) and $k = -1$ (October 2022)
- SE clustered at occupation

Why Poisson? Counts with some zeros; $\log(0 + x)$ biased. Q3 (median) reference in backup.

Q3 reference

Cell-level event-study grid (Q1 reference)



Q2–Q5 vs. Q1, by decade age group, private sector; reference October 2022 ($k = -1$). At 21–30, Q5 is noisy and ends near zero; 31–40 rises to about $+0.15$; 41–50 reaches about -0.08 by mid-2025 and ends near -0.03; 51–60 stays near zero.

Difference-in-differences: Q5 vs. Q1, post-ChatGPT average

	21–30	31–40	41–50	51–60
Q5 × Post	0.020	0.081***	−0.015	0.010
	(0.023)	(0.019)	(0.017)	(0.017)
Occupations	391	393	393	392

- Stacked triple difference, Q5 × Post × young (21–30 vs. 31–60): −0.009 (0.016), n.s.
- No larger post-ChatGPT decline for young workers in the most exposed quintile.

Poisson, occupation and month FE, SE clustered at occupation. *** $p < 0.01$.

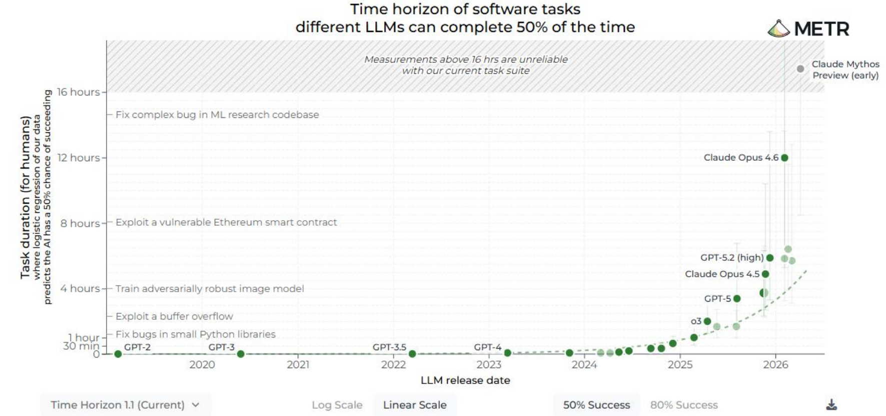
Pre-trends are not parallel. How much does it matter?

- Joint tests reject parallel pre-trends in every age group; the Q5 pre-period path moves 0.11 to 0.18 log points, as large as the post-2022 movements.
- HonestDiD (Rambachan & Roth 2023): post-period average Q5 vs. Q1, ages 21–30.

Restriction	Robust 95% CI
Original ($\bar{M} = 0$, parallel trends)	[-0.026, +0.065]
$\bar{M} = 0.5$	[-0.234, +0.314]
$\bar{M} = 1.0$	[-0.515, +0.582]
$\bar{M} = 1.5$	[-0.783, +0.863]
$\bar{M} = 2.0$	[-1.050, +1.130]
Breakdown $\bar{M}$	0.00

The conventional interval already includes zero, so there is no gradient to overturn (breakdown $\bar{M} = 0$). But the null is tight only under parallel trends: the robust intervals do not rule out large effects in either direction.

AI capability is accelerating



Length of software tasks an LLM completes at 50 % success has roughly doubled every ∼7 months since GPT-2. By 2026, the frontier handles multi-hour tasks. Source: Model Evaluation & Threat Research (METR).

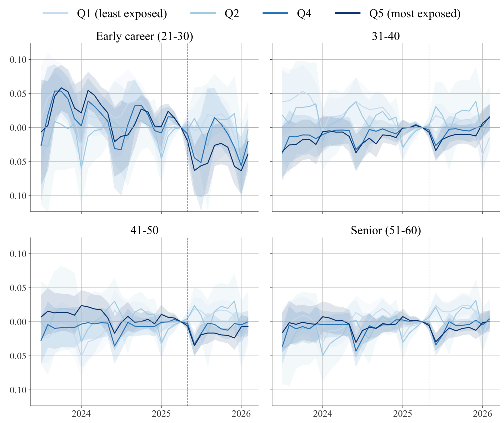
Why agentic AI is a separate window

Mechanism. Pre-agentic AI: AI helps with particular steps of a production chain, humans verify each step. Agentic AI: humans verify only the final output of a chain (Demirer, Horton, Immorlica & Lucier 2026).

When did this happen? Somewhat arbitrary, user cost falls gradually.

- **Early 2024.** Devin demo (Mar 2024), Claude 3 Opus (Mar 2024), standardised function-calling APIs. Chaining technically possible but demo-grade.
- **May 2025.** OpenAI Codex (16 May 2025) and Anthropic Claude 4 / Claude Code (22 May 2025) make multi-step coding agents production-grade.

Re-anchor to agentic AI (April 2025)



Same Poisson DiD, re-anchored to April 2025 (last pre-agentic month). Pre-period 2023m1–2025m4, post 2025m5–2026m2. Cell-level, private sector. Q3 reference, estimated on data through February 2026.

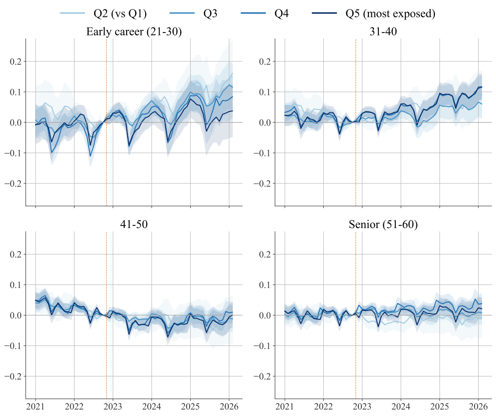
Validation: individual-level data

Validation: within-firm reallocation

- The cell-level gradient could reflect compositional shifts across firms rather than within-firm reallocation.
- We re-estimate on individual-level register data with firm fixed effects, as in Brynjolfsson–Chandar–Chen.
- Individual-level data is a snapshot; we use it to validate the cell-level pattern, not for time-tracking.

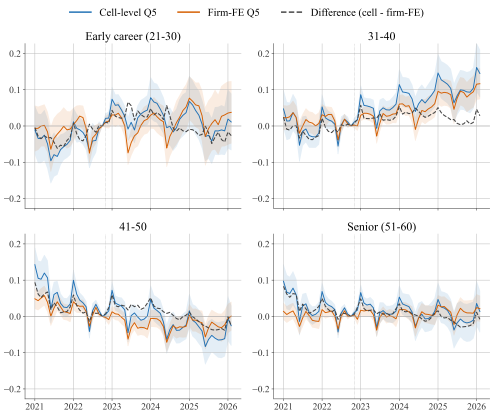
Specification. Poisson PPML on firm $\times$ quintile $\times$ month cells, per decade age group, with firm $\times$ quintile and firm $\times$ month fixed effects. Identification from within-firm reallocation across exposure quintiles. Sample: private *foretak* with ≥ 20 workers aged 21–60, 2021m1–2026m2; about 22,000 firms covering 66–72 % of private paid employment. SE clustered at *foretak*.

Within-firm event study (Q1 reference)



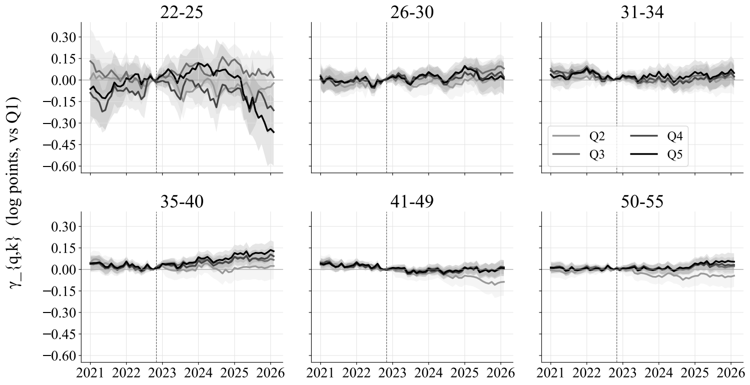
Q2–Q5 vs. Q1 by decade age group. At 21–30, Q5 is noisy and ends slightly positive; 31–40 rises to about +0.12; 41–50 drifts negative through 2025 and recovers to near zero; 51–60 stays flat. Collapsed Q5 × Post: +0.013 (n.s.), +0.050***, −0.021*, +0.007.

Cell vs. within-firm: same story



Nested reconciliation of the Q5 estimates: switching data source moves nothing (≤ 0.003 in any age group); the ≥ 20-worker restriction is small; the largest single step is cell to firm-FE, mainly at 31–40 ($+0.084 \rightarrow +0.050$). Signs unchanged throughout.

Replicating Brynjolfsson et al. (ages 22–25)



Balanced panel of full-time private-sector workers, six age bins. For ages 22–25 the average post-ChatGPT Q5 vs. Q1 gap is small ($\sim$ 3 log points), but Q5 falls below Q1 from spring 2025, widening to about 35 log points by February 2026, more than the 15 reported for the US. Late, imprecise, and on rejected pre-trends.

Cross-country comparison

Cross-country: where Norway lands

	Descriptive (case studies)	Formal estimation (young, Q5 vs. Q1)
US	SW developers 22–25: −20 %	−15 log pts (firm FE)
Sweden	≈ −44 % raw, top quartile	−5.5 % within employer by 2025H1
Denmark	aggregate early-career decline	null; rules out > 2 % earnings effect
Finland	null	null
Norway	SW developers 21–30: −18 %	≈ 0 at 21–30; 22–25 diverges only from spring 2025

- On case studies, Norway looks like the US and Sweden.
- On systematic full-distribution evidence, Norway looks like Denmark and Finland.
- Cross-country differences may reflect differently confounded environments (rate hikes, post-COVID tech correction) rather than different AI effects.

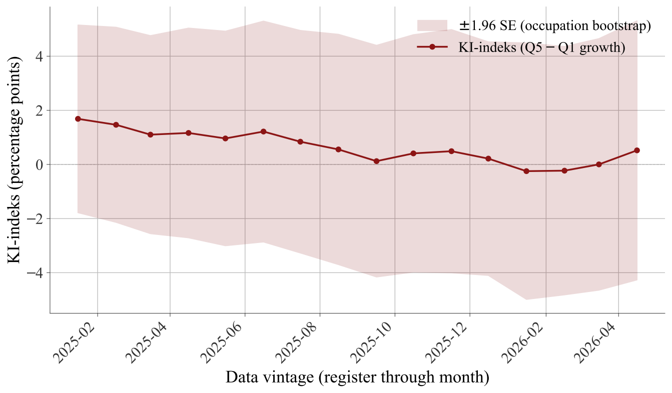
The dashboard: kiindeksen.no

Employment change by quintile since ChatGPT

	Q1	Q2	Q3	Q4	Q5
Average change (%)	-0.63	+2.94	+0.95	+2.86	-0.09
Difference vs. Q1 (%)	—	+3.57	+1.59	+3.49	+0.54
		(5.36)	(2.67)	(2.15)	(2.42)
Occupations			374		

Change from October 2022 to the February–April 2026 average; seasonally adjusted private-sector headcount, ages 21–60, occupations weighted by October 2022 employment (Yagan 2019 style). Most exposed $-0.09\,\%$ vs. least exposed $-0.63\,\%$: a $+0.54$ pp double difference (SE 2.42), not significant. Robust SEs across occupations in parentheses.

kiindeksen.no: a live index



The headline "KI-indeks": Q5 vs. Q1 employment growth, most recent three months vs. October 2022. Across expanding data vintages it drifts from +1.7 to −0.2 and back to +0.5 pp (April 2026). Bootstrap SE of 2 to 2.5 pp throughout: never statistically distinguishable from zero. A genuine change = a sustained move beyond the band.

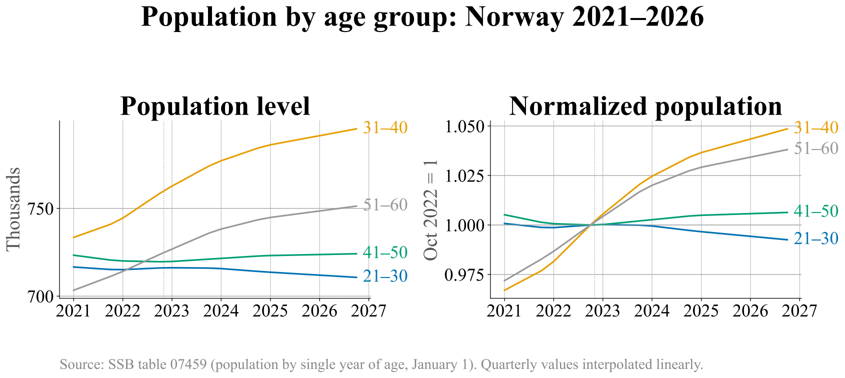
Conclusion

Takeaway

- **No aggregate job crisis.** Private-sector employment is growing; the employment rate is stable.
- **Full distribution.** Since October 2022, employment fell 0.1 % in the most exposed occupations against 0.6 % in the least exposed: a +0.5 pp gap, well within the ∼2.5 pp bootstrap SE.
- **The canary is real but does not generalize.** Young workers in software development and customer service fell after ChatGPT; across full exposure quintiles, Q5 did not fall behind Q1 at ages 21–30. Pre-trends warrant caution in both directions.
- **Cross-country.** Norway resembles the US and Sweden on case studies, Denmark and Finland on systematic evidence.
- **Monitoring continues** monthly at kiindeksen.no (∼3-month data lag). All results private sector.

Backup

Population and cohort adjustment



- Per-capita normalisation (divide by SSB resident population in the same age cell).
- Composition adjustment fixes the within-bin age distribution at 2021-Q1.
- The age-by-quintile pattern is unchanged after both adjustments.

Lindenlaub et al. (2026), Fig. 1: AI adoption vs. four exposure measures

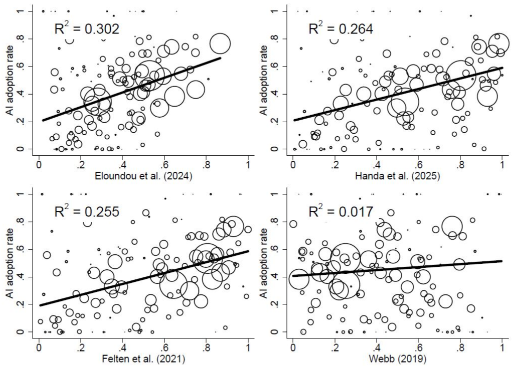
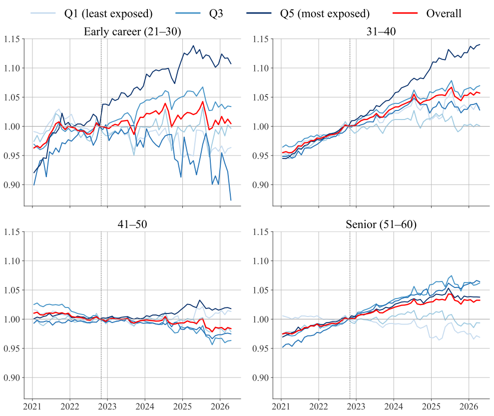


Figure 1: Relation between Actual Adoption by Workers and AI Exposure Measures

Each panel regresses worker-reported AI adoption on an exposure measure. Eloundou et al. has $R^2 = 0.30$, ahead of Handa (0.26), Felten (0.26), and Webb (0.02).

Back

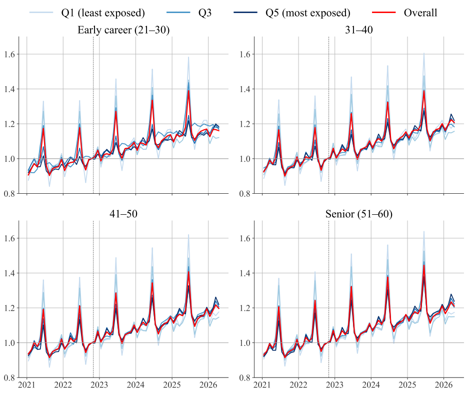
Public sector: no exposure gradient



Same indexed series as the private-sector slide, public sector, seasonally adjusted. No exposure gradient at any age.

Back

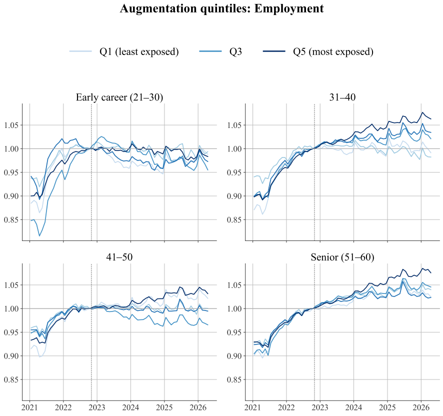
Cash earnings: no divergence



Cash earnings (*kontantlønn*) by quintile and age, private sector. Seasonal patterns dominate (holiday pay, December bonuses): a tentative null, consistent with adjustment on the hiring margin under two-tier bargaining.

Back

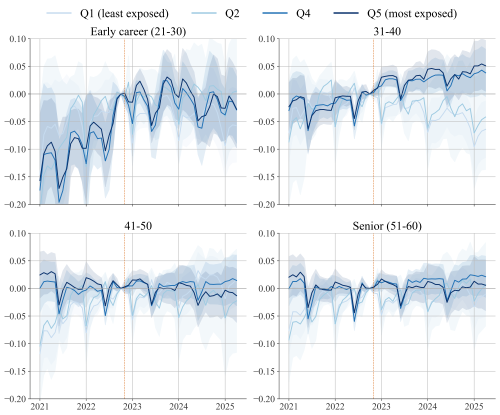
Handa augmentation: no gradient



Quintiles ranked by **augmentation** share of Claude usage, seasonally adjusted. Gaps are less pronounced than for automation; the least exposed are sometimes on par with the most exposed.

Back

Cell-level event study, Q3 reference



Same cell-level Poisson event study, plotted against Q3 (median exposure) instead of Q1. Q3 avoids Q1's winter-construction seasonality; the qualitative pattern is unchanged.

Back

Backup: mapping worked examples

- Five further one-to-one occupations, with full task tables: carpenters, secondary teachers, lawyers, economists, ICT operations technicians.
- A many-to-one example (STYRK 1112) and the O*NET data structure.

Back

Other one-to-one mappings (large Norwegian occupations)

STYRK	Norwegian title	SOC 2010 (US title)	employed	β_{occ}	Q
7115	Tømrere og snekkere	47-2031 Carpenters	47,106	0.144	2
2330	Lektorer (videregående skole)	25-2031 Secondary teachers	31,155	0.333	3
2611	Jurister og advokater	23-1011 Lawyers	11,808	0.425	4
2631	Forskere/rådgivere, samf.økonomi	19-3011 Economists	1,282	0.553	5
3511	Driftsteknikere, IKT	43-9011 Computer Operators	16,517	0.736	5

All five are clean one-to-one in the BLS crosswalk. Exposure spans Q2 to Q5: carpenters at the bottom, ICT operations technicians at the top. β_{occ} is taken from the published Core-weighted occupation file, so it can differ slightly from an unweighted task mean. SOC titles do not always match the Norwegian title literally (e.g. "Computer Operators" vs. "Driftsteknikere, IKT").

Worked example: Tømrere og snekkere (STYRK 7115)

O*NET task		SOC 2018		SOC 2010		ISCO-08
29 tasks	$\xrightarrow{\textit{average}}$	47-2031	$\xrightarrow{1\to1}$	47-2031	$\xrightarrow{1\to1}$	7115
$E_0/E_1/E_2$		Carpenters		Carpenters		Carpenters

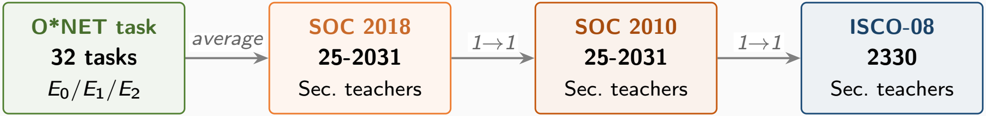
STYRK 7115 is Tømrere og snekkere. Same 6-digit SOC in both vintages, one O*NET specialty, 29 tasks. Mostly E_0: physical work dominates.

Task-level scoring: Carpenters (SOC 47-2031, 29 tasks)

O*NET task	E	β	O*NET task	E	β
Follow safety rules and regulations	E_0	0.0	Dig or direct digging of post holes	E_0	0.0
Measure and mark cutting lines on materials	E_0	0.0	Cover subfloors with building paper	E_0	0.0
Assemble and fasten materials	E_0	0.0	Construct forms or chutes for concrete	E_0	0.0
Study specifications in blueprints	E_2	0.5	Arrange for subcontractors	E_2	0.5
Shape or cut materials to measurements	E_0	0.0	Build or repair cabinets, doors, floors	E_0	0.0
Verify trueness with plumb bob and level	E_0	0.0	Finish surfaces of woodwork or wallboard	E_0	0.0
Inspect ceiling/floor tile, wall coverings	E_2	0.5	Select and order lumber and materials	E_2	0.5
Erect scaffolding or ladders	E_0	0.0	Work with or remove hazardous material	E_0	0.0
Install windows, frames, fixtures	E_0	0.0	Fill cracks or defects in plaster	E_0	0.0
Maintain records, present written reports	E_1	1.0	Prepare cost estimates for clients	E_2	0.5
Remove damaged or defective parts	E_0	0.0	Perform minor plumbing or concrete work	E_0	0.0
Maintain job records, schedule work crew	E_2	0.5	Apply shock-absorbing or sound-deadening	E_0	0.0
Anchor and brace forms and structures	E_0	0.0	Examine structural timbers for decay	E_0	0.0
Bore boltholes in timber/masonry/concrete	E_0	0.0	Build sleds from logs and timbers	E_0	0.0
Install rough door/window frames, subflooring	E_0	0.0			

Counts: 22 E_0 1 E_1 6 E_2 **SOC β (mean of 29 tasks): 0.138 (Q2)**

Worked example: Lektorer videregående (STYRK 2330)



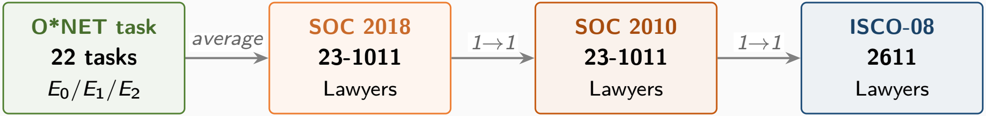
STYRK 2330 is Lektorer mv. (videregående skole). One O*NET specialty, 32 tasks. Most are E_2 (LLM with software is enough) and reflect classroom and administrative work.

Task-level scoring: Secondary teachers (SOC 25-2031, 32 tasks)

O*NET task	E	β	O*NET task	E	β
Instruct through lectures, discussions	E_2	0.5	Plan and conduct field trips	E_0	0.0
Establish rules of student behavior	E_0	0.0	Sponsor extracurricular activities	E_0	0.0
Prepare materials and classrooms	E_2	0.5	Perform administrative duties	E_2	0.5
Adapt teaching to varying abilities	E_2	0.5	Prepare and implement remedial programs	E_2	0.5
Observe and evaluate students' performance	E_2	0.5	Attend professional meetings and workshops	E_0	0.0
Establish clear objectives for lessons	E_2	0.5	Use computers and audiovisual aids	E_2	0.5
Prepare students for later grades	E_2	0.5	Select, store, order, or use class materials	E_2	0.5
Prepare, administer, and grade tests	E_2	0.5	Collaborate with other teachers	E_2	0.5
Maintain accurate, complete records	E_2	0.5	Plan and supervise projects, field studies	E_2	0.5
Enforce attendance and progression rules	E_0	0.0	Meet with professionals on programs	E_0	0.0
Instruct and monitor use of equipment	E_0	0.0	Guide and counsel students	E_2	0.5
Prepare content for diverse-needs students	E_2	0.5	Confer with parents on progress	E_0	0.0
Assign and grade class work and homework	E_2	0.5	Meet with administrators on instruction	E_2	0.5
Prepare reports on student progress	E_2	0.5	Prepare for assigned classes	E_2	0.5
Keep informed about developments in education	E_1	1.0	Provide disabled students with devices	E_2	0.5
Prepare objectives and outlines	E_2	0.5	Administer standardized tests	E_0	0.0

Counts: 12 E_0 1 E_1 19 E_2 **SOC β (mean of 32 tasks):** **0.328 (Q3)**

Worked example: Jurister og advokater (STYRK 2611)



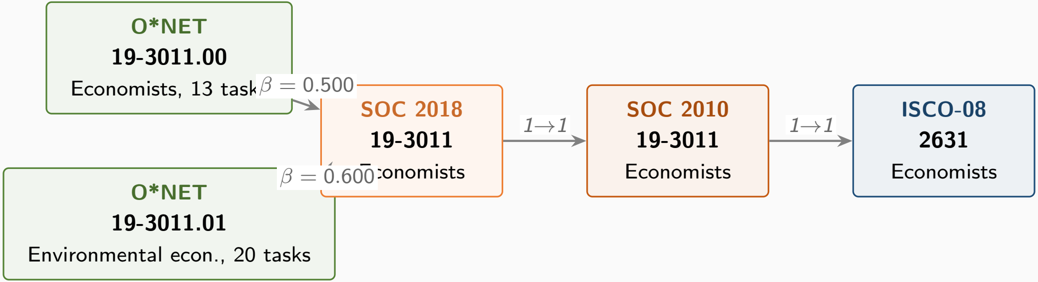
STYRK 2611 is Jurister og advokater. One O*NET specialty, 22 tasks. Nearly all tasks are E_2: text-heavy work where LLM plus software can substitute.

Task-level scoring: Lawyers (SOC 23-1011, 22 tasks)

O*NET task	E	β	O*NET task	E	β
Advise clients on legal rights	E_2	0.5	Confer with colleagues with specialties	E_0	0.0
Represent clients in court or in hearings	E_0	0.0	Examine legal data for advisability	E_2	0.5
Select jurors, argue motions, meet judges	E_0	0.0	Search and examine public records	E_2	0.5
Study Constitution, statutes, decisions	E_2	0.5	Act as trustee, guardian, or executor	E_2	0.5
Prepare legal briefs and opinions	E_2	0.5	Help establish company legal policies	E_2	0.5
Interpret laws, rulings, regulations	E_2	0.5	Negotiate settlements of civil disputes	E_2	0.5
Prepare and draft legal documents	E_2	0.5	Probate wills, represent estate executors	E_2	0.5
Present and summarize cases to judges	E_2	0.5	Work in environmental, public health law	E_2	0.5
Gather evidence by interviewing clients	E_2	0.5	Perform administrative and management	E_2	0.5
Analyze the probable outcomes of cases	E_2	0.5	Supervise legal assistants	E_2	0.5
Evaluate findings; develop defense strategy	E_2	0.5	Provide pro bono services	E_2	0.5

Counts: 3 E_0 0 E_1 19 E_2 SOC β (mean of 22 tasks): 0.432 (Q4)

Worked example: Forskere/rådgivere, samf.økonomi (STYRK 2631)



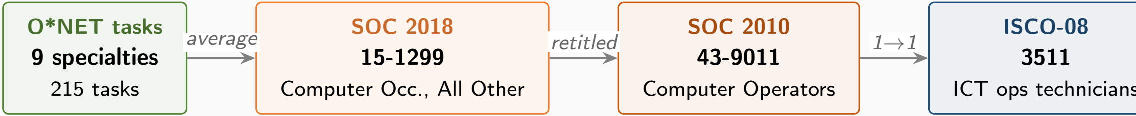
Two O*NET specialties roll up to one SOC 2018 code. The SOC-level β averages the two specialty means: $(0.500 + 0.600)/2 = 0.550$, regardless of how many tasks each specialty contains.

Task-level scoring: Economists (SOC 19-3011, 33 tasks across 2 specialties)

19-3011.00 Economists (13 tasks)			**19-3011.01 Env. Economists (20 tasks)**		
Study economic and statistical data	E_2	0.5	Prepare and deliver presentations	E_2	0.5
Conduct research on economic issues	E_2	0.5	Develop programs or policy recommendations	E_2	0.5
Compile, analyze, report data	E_2	0.5	Develop economic models and forecasts	E_2	0.5
Supervise research projects	E_2	0.5	Demonstrate economic benefits of policies	E_2	0.5
Teach theories and methods of economics	E_2	0.5	Conduct research on environmental relations	E_2	0.5
Study impacts of public policies	E_2	0.5	Perform complex mathematical analyses	E_1	1.0
Formulate recommendations and plans	E_2	0.5	Write social, legal, economic impact reports	E_1	1.0
Explain economic impact of policies	E_2	0.5	Teach courses in environmental economics	E_2	0.5
Provide advice and consultation	E_2	0.5	Develop programs to promote sustainability	E_2	0.5
Forecast production and consumption	E_2	0.5	Develop systems for analyzing data	E_2	0.5
Develop economic guidelines and standards	E_2	0.5	Write research proposals and grants	E_1	1.0
Testify at regulatory or legislative hearings	E_2	0.5	Examine exhaustibility of natural resources	E_2	0.5
Provide litigation support	E_2	0.5	Monitor or analyze market and env. trends	E_2	0.5
Write technical documents or articles	E_1	1.0	Develop environmental research project plans	E_2	0.5
			Conduct research on economic, env. topics	E_2	0.5
			Collect and analyze data on env. impact	E_2	0.5
			Assess costs and benefits of activities	E_2	0.5
			Identify environmentally friendly business	E_2	0.5
			Interpret indicators of ecosystem health	E_2	0.5
			Collaborate with researchers across fields	E_2	0.5
Specialty mean: 0.500			**Specialty mean:** 0.600		

SOC $\beta = (0.500 + 0.600)/2 =$ **0.550 (Q5)**

Worked example: Driftsteknikere, IKT (STYRK 3511)

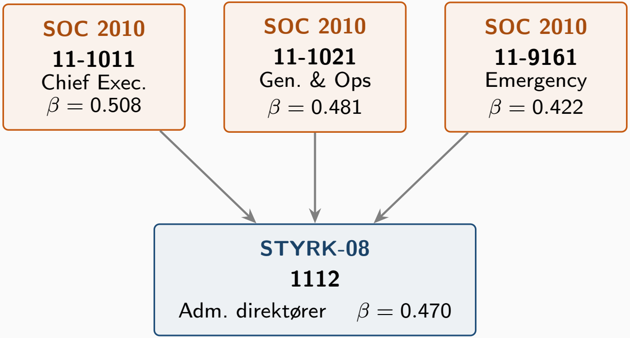


The SOC 2010 → 2018 re-coding turned "Computer Operators" into the catch-all "Computer Occupations, All Other" (15-1299), which contains nine sub-specialties. The Eloundou et al. pipeline averages those nine specialty means; tasks are not aggregated directly.

Specialty-level scoring: Computer Occupations, All Other (15-1299)

O*NET-SOC	Title	N tasks	specialty β
15-1299.01	Web Administrators	23	0.913
15-1299.02	Geographic Information Systems Technologists	27	0.593
15-1299.03	Document Management Specialists	18	0.611
15-1299.04	Penetration Testers	22	0.795
15-1299.05	Information Security Engineers	20	0.675
15-1299.06	Digital Forensics Analysts	20	0.725
15-1299.07	Blockchain Engineers	17	0.971
15-1299.08	Computer Systems Engineers/Architects	28	0.768
15-1299.09	Information Technology Project Managers	21	0.571
SOC β (mean of 9 specialties): 0.736 (Q5)			

Worked example: many-to-one mapping



- Three SOC 2010 codes map to ISCO 1112 in the BLS crosswalk
- STYRK β is the unweighted mean of contributing SOC-level β
- 57.7% of Eloundou et al. STYRK codes have at least one contributor flagged as a partial match in the BLS file

O*NET: the US occupation database underneath three of the four measures

- Maintained by the US Department of Labor; covers ∼1,000 occupations on SOC 2018 codes (O*NET-SOC 2019 taxonomy)
- Each occupation has structured content:
  - **Tasks:** discrete activities (∼ 19,000 across all occupations; Pharmacists: 21 tasks, e.g. "review prescriptions for accuracy"). Tagged *Core* or *Supplemental*; no numerical weight.
  - **Abilities:** 52 underlying attributes (e.g. Deductive Reasoning, Manual Dexterity). Scored on prevalence and importance per occupation.
  - Also: skills, knowledge, work activities, work context (not used by the four measures here).
- Eloundou et al. and Handa operate on **tasks**; Felten operates on **abilities**; Anthropic 2026 operates at the occupation level.